13. If you seal the hole of the ink tube of a pen, the pen will no longer functions why?
14. When you walk along a road and a truck passes you at high speed. Why do you feel pulled towards the truck? Explain
15. Desert travelers keep water in a canvas bag. Some water seeps through the bag and evaporates. How does this cool the water inside the bag?
16. When you first step out of the shower. You feel cold. But as soon as you are dry. You feel warmer. Why?
17. A cold block of metal feels colder than a block of wood at the same temperature. Why?
18. It is well known that a potato bakes faster if a large nail is stuck through it. Why?
19. What is wrong with the statement, "given any two bodies, the one with the higher temperature contains more heat?"
20. If you want to drink a cup of hot coffee within a reasonable short time, what should you do with it?
21. Why can you get a more severe burn from steam at 100^0C than from water at 100c?
22. If you hold water in paper cup over a flame, you can boil the water without burning the cup. How is this possible?

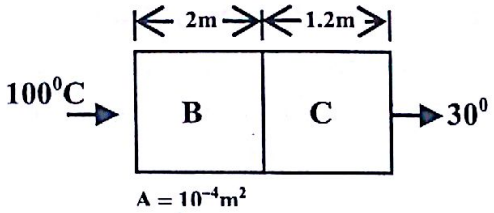
Unit 8: Problems

23. A load of 2kg is applied to the ends of a wire 4m long, and produces an extension of 0.24mm if the diameter of the wire is 2mm find?
    a) The stress on the wire
    b) The strain it produces c) The young's modulus
24. a) What load in kilograms must hung form a steel wire 6m long and radius 0.8mm to produce an extension of 1mm? Young's modulus for steel is 2×10^{11}pa
    b) What is the strain energy?
25. At the top of a mountain a barometer reads 71cm of Hg and at the bottom it reads 76cm of Hg. What is the height of the mountain
26. Water is flowing through a closed pipe system at one point the speed of water is 2.8m/s, at a point 4cm higher than the first the speed is 4.2m/s. If the pressure at the lower point is 84kpa, what is the pressure at the upper point?
27. A horizontal pipe 10cm in diameter in reduced to a pipe of diameter 5cm. If the pressure in the larger pipe is 150kpa, and

30kpa in the smaller, what is the speed of water in the larger pipe?

28. A box shaped piece of material has a top area of $40cm^2$ and a height of 8cm, when a shearing force of 0.8N is applied to the upper surface, the upper surface displaced by 8mm relative to the bottom. Find the
    a) shear stress b) shear strain c) shear modulus
29. The speed of blood in a major artery of radius 0.4cm is 0.04cm/s. If the artery is divided in to capillaries of radius 0.1cm, what is the speed of the blood through each capillary?
30. If a force of 50N is applied on to a piston of area 0.5 m^2. What load of can on the other piston of area $5m^2$ in a hydraulic machine can be lifted?
31. How much heat is required to convert 20g of ice at 0^0c to water at 20^0C.? L_f= 80 cal/g, C= 1 cal/gC^0
32. The operating temperature of a tungsten filament in an incandescent lamp is 2450k and its emissivity is 0.3. What is the surface area of the filament of a 25w lamp?
33. A sphere of 3cm in radius acts like a black body. It is in equilibrium with its surrounding and absorbs 30,000 w of power radiated to it from the surrounding. Calculate the temperature of the sphere
34. The weight of a block having density $5\times10^3 kg/m^3$ is 1.5N in air. When the block is completely immersed in a certain liquid its weight is found to be 1.2N. What is the density of the liquid?
35. Through a pipe, 15cm in diameter water is pumped from the river outside of Addis Ababa to one of the sub cities of Addis Ababa. The river is at an elevation of 564cm and the sub. City is at an elevation of 2096m. What is the minimum pressure at which the water must be pumped if it is to arrive at the sub city?
36. By which mechanism of heat Transfer you get heat by holding a piece of metal?
37. A 0.05kg metal is heated to 200°c and then dropped in to a beaker of 0.4kg water initially at 20°C. If the final temperature of the mixed system is 22.5°C, what is the specific heat of the metal?
38. What mass of steam initially at 1300c is needed to warm 200g of water in a 100g glass container from 20^0C to 50^0C? $C_s = 2.01 \times 10^3$ J/kg ^{0}C, $L_v = 2.26 \times 10^6$, $C_G = 837$J/kg^{0}C

39. A piece of aluminum is suspended from a string and then completely immersed in a container of water. The mass of the aluminum is 1kg and its density is $2.7 \times 10^3 kg/m^3$. Calculate the tension in the string before and after the aluminum is immersed
40. An ice cube floats in a glass of water what fraction of the cube lies above the water level?
41. A15kg gold block is raised from an ocean using a cable. What is the tension in the cable when the block is completely immersed
42. A storage tank 12m deep is filled with water. The top of the tank is open to the air. What is the *absolute* pressure at the bottom of the tank?
43. What is the total pressure when the static pressure is 9pa and the dynamic pressure is 3pa
44. Work out the rate of flow of heat through the bar. Let the thermal conductivity of B = 450W/mk while for C = 150W/mk



45. A 0.5kg mass is hung from the end of a wire 1.5m long and 0.3mm diameter. If young's modulus for it is 1×10^{11}pa. Calculate the extremism and strain energy
46. Find the height of a column of methyl iodide in a soda lime glass tube of radius 20mm, the surface tension is 0.26, the contact angle is 29^0, the density is 2.28
47. Find the force on a ball bearing of radius 5×10^{-3}m falling through a liquid of viscosity 0.985 at a velocity of 0.35m/s.

Unit 8: Explanations to Questions

1. First find mass, $m = \frac{mg}{g}$, then buoyant force = W_{air} - W_{water}. But $B_F = \rho_w Vg = W_{air} - W_{water}$. You get V, and $\rho = \frac{m}{v}$, compare it with ρ_{Gold}
2. In the lake because density of lake water (fresh) is less than density of Ocean water (salty)
3. By placing the block with the iron on top side
4. No change of level
5. No! it doesn't work for an ideal fluid where the speed is steady

6. Aircraft engines develop less power in thin air and the pressure difference is very small
7. The hull of the ship is full of air and the density of air is about one thousandth the density of water hence the total weight of the ship is less than the weight of an equal volume of water
8. The fast moving air causes the external pressure to fall. The pressure difference between the inside and outside results in an upwared force on the roof.
9. Atmospheric pressure accompanied by capillary action.
10. Because water is much less dense than mercury for 1atm.
    h $= \frac{P}{\rho g} = \frac{101000}{(1000)(10)} \approx 10m$. We need 10m column $\rightarrow$ not manageable
11. After exhaling
12. Because air is less dense than equal volume of water
13. The outside pressure will be large than the inside pressure
14. The air pressure between you and the train decreases due to high speed thus the outside pressure push you to the train.
15. As water evaporates from the surfaces it takes heat to the atmosphere
16. At first heat transfers from our body to the surrounding and then thermal equilibrium is reached
17. Because, metal is good conductor of heat but wood it not.
18. Because the nail transports thermal energy
19. Heat content depends on mass, specific heat capacity and temperature of a body not only on temperature
20. It is good to insert a metal spoon in to the cup so that it will conduct the heat from the coffee to the outside.
21. Steam at 100^0C do have a hidden heat called latent heat.
22. Because paper is isolative and thin enough to act as mediator.

Unit 8: Solution to problems

23. a) stress $= \frac{F}{A} = \frac{mg}{\pi(1\times10^{-3})^2} = \frac{20}{(3.14)(10^{-6})} = 6.4Mpa$

    b) strain $= \frac{X}{L} = \frac{0.24\times10^{-3}}{4} = 0.00006$

    c) E $= \frac{FL}{XA} = 1.06 \times 10^{11}Pa$
24. a) $E = \frac{FL}{XA} \Rightarrow F = \frac{EXA}{L} = 6.072kg$

    b) strain energy $= ½FX = ½ \times 67.02 \times 1 \times 10^{-3} = 0.034J$
25. 527m

26. $\frac{P_1}{\rho} + gh_1 + \frac{V_1^2}{2} = \frac{P_2}{\rho} + gh_2 + \frac{V_2^2}{2}$, $h_1 = 0$, $h_2 = 0.04$m
$\Rightarrow P_2 = 39.1$kpa

27. $\frac{P_1}{\rho} + \frac{V_1^2}{2} = \frac{P_2}{\rho} + \frac{V_2^2}{2}$, $h_1 = h_2 = 0$ and $A_1V_1 = A_2V_2 \Rightarrow V_2 = 4V_1$
$\Rightarrow 150 + \frac{V_1^2}{2} = 30 + \frac{16V_2^2}{2} \Rightarrow V_1 = 16$m/s

28. a) $\frac{F}{A} = 200pa$ b) $\frac{X}{L} = \frac{8}{80} = 0.1$ c) $\frac{200}{0.1} = 2000pa$

29. $A_1V_1 = A_2V_2 \Rightarrow V_2 = 0.64$cm/s

30. $\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = \frac{A_2}{A_1}F_1 = 500N$

31. $Q = mLf + mC\Delta T = (20)(80) + (20)(1)(20) = 2000$Cal

32. $P = \sigma eAT^4 \Rightarrow A = \frac{25}{5.67} \times 10^{-8} \times 0.3 \times (2450)^4 = 4.08 \times 10^{-5}m^2$

33. $P = \sigma eAT^4$, $e = 1$, $\sigma = 5.67 \times 10^{-8} \Rightarrow T = 2600$k

34. $F_b = W_{air} - W_{water} = 1.5 - 1.2 = 0.3$. $F_b = \rho_f V_f g$. but $V_f = V_0 = \frac{m_0}{\rho_0}$
$\Rightarrow 0.3 = \rho_f = 10^3$kg/m^3

35. $\Delta P = \rho g \Delta h = (1000)(10)(2096 - 564) = 15.32$Mpa

36. conduction

37. $Q_{lost} = Q_{gaind} \Rightarrow mC(T_i - T) = m_wC_w(T - T_i) \Rightarrow C = 453$J/kg^{0}C

38. $Q_1 = m_sC_s\Delta T$, $Q_2 = m_sL_v$, $Q_3 = m_sC_w\Delta T$
$Q_1 + Q_2 + Q_3 = (0.2 \times 4.19 \times 10^3 \times 30) + (0.1 \times 837 \times 30) \Rightarrow m_s = 10.9$g

39. $T_1 = mg = 9.8N$, $T_2 = mg - F_B = mg - \rho_w\left(\frac{m}{\rho}\right)g = 6.2N$

40. $F_B = \rho_w Vg$, V volume under water. Since it floats;
$W = F_B = \rho_c V_c g = \rho_w Vg \Rightarrow \frac{V}{V_C} = \frac{\rho_w}{\rho_c} \Rightarrow f_{rc} = 1 - \frac{\rho_c}{\rho_w} = 8.3\%$

41. $T = mg - B_F$, $B_F = \rho_w Vg = (1.03 \times 10^3)\left(\frac{m}{\rho_{gold}}\right)(10) \Rightarrow T = 139N$

42. $P = P_0 + \rho gh = (1.01 \times 10^5) + (1000)(10)(12) = 2.21 \times 10^5$Pa

43. $P_0 = p + q = 9 + 3 = 12$Pa

44. $450 \times A \times (100 - \theta)/2 = 150 \times A(\theta - 30)/1.2 \Rightarrow \theta = 75$
for bar B. $\frac{Q}{t} = 450 \times 10^{-4}(100 - 75)/2 = 0.5625$W

45. $x = \frac{FL}{EA} = 0.001m$, energy = ½FX = 0.0025J

46. $h = \frac{2f\cos\theta}{\rho gr} = 1.02m$

47. $F = 6\pi\eta rv = 6 \times 3.14 \times 0.985 \times 5 \times 10^{-3} \times 0.35 = 0.032$N

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